

1. Purpose

To provide some explanations using non-Azure terms to explain questions that we often receive.

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3. Reference

3.1. Azure Fraud Access

When an Azure environment is compromised and fraud occurs, the end user will often say they should not be responsible for the charges because Microsoft provides the service.

In some cases, the customer may be confused by the Fair Credit Billing Act (FCBA) which only applies to credit card companies.



We can think of this situation as though Microsoft were a home builder, and the end user customer is the homeowner.

- Microsoft builds a home with windows and a door. The windows have locks and the door has a doorknob with a lock.
- The doorknob is analogous to a password. If the doorknob is not locked, a burglar could easily enter the house. Should the builder be responsible if the homeowner did not lock the doorknob?

- A doorknob may have been sufficient in the 1940's; however, as burglars got more proficient, we had to upgrade to deadbolts. Would the homeowner expect the builder to come to the home in 2024 and install deadbolts? Probably not, however, Microsoft has provided MFA which is an upgrade to security like a deadbolt.
- If the builder did install deadbolts and the homeowner did not use them, should the builder be responsible if the homeowner did not lock them? If the customer did not use MFA, should Microsoft be responsible?
- What if the homeowner tried to lock the deadbolt but did not close the door first? There are many resources that explain how to close a door and lock a deadbolt. We would probably not fault the builder. We have seen cases where customers have tried to implement MFA but did it incorrectly. There are plenty of resources about setting up MFA. Is Microsoft responsible if the customer never looked for resources?
- There are cases where the homeowner may have locked the door but forgot to close the window. A burglar then climbs through the window. Again, should the builder be at fault? Why didn't the builder post a guard outside the house to watch it?
 - The customer may have enabled MFA for some services but not for others. This oversight has led to compromised environments. There are resources and templates available to help prevent this.
 - The Sentinel service and SOCaaS services could replace the guard watching for unexpected vulnerabilities.
 - Customers routinely leave RDP ports open which can lead to compromises.

Microsoft has provided tools such as MFA to customers at no cost. There are many resources available from Microsoft and TD SYNnex to help the customer understand how to secure their environments.

Just like the homeowner is responsible for locking their door, the customer is responsible for securing their environment.

3.2. Microsoft Sentinel Monitoring

Microsoft Sentinel is a Security Information and Event Management (SIEM) solution designed to enhance threat monitoring and response.



To illustrate this, we can think of it with this analogy.

- Imagine your organization's security as a house with a door and multiple windows. Each entry point represents a potential vulnerability.

- Locking the door and securing each window individually is like traditional security practices. You're vigilant, but it requires splitting your attention across all potential threats.
- Microsoft Sentinel acts as an integrated alarm system. It allows you to monitor all windows and the door simultaneously. Think of it as having a single monitor that consolidates all risks, making it easier to detect and respond to threats.
- However, being alerted is just the first step. Can you respond effectively at all hours? Here's where a Security Operations Center as a Service (SOCaaS) comes in. SOCaaS provides expert analysts who monitor alerts 24/7. Remember, having Sentinel running without active monitoring can create a false sense of security. It's crucial to combine technology with human expertise for comprehensive threat management.

3.3. Virtual Machine Charges

When a virtual machine is Stopped in Azure it can be compared to turning off the power to the virtual machine on-prem. The charges will still accrue.

When a virtual machine is Stopped (Deallocated) in Azure it can be compared to removing it from the data rack and returning it to the manufacturer. The charges stop.

Another way to think about the reason that Microsoft charges for the Stopped state is to consider the case of a car rental company.

3.3.1. Creating

- If I rent a car from a rental company (creating a VM), the rental company cannot rent the car to another consumer. I have exclusive use of the car, so they charge me for the time I have the car.

3.3.2. Running

- If I am driving the car (VM in a running state), I am again using the car, so I am charged.

3.3.3. Stopped

- But what if I park the car in my driveway (VM in stopped state)? I am not using the car or adding mileage to it. The rental company will still charge you for the car because they cannot rent it to another consumer.
- Parking the car in your driveway can also be compared to a Reserved Instance. You agree to pay Microsoft even if you do not use the car.

3.3.4. Stopped (Deallocated)

- If you return the car to the rental company (Stopped – Deallocated state) then they will stop charging you. They can rent it to another consumer but even if they do not have someone to rent it, they will not charge you for the car sitting on the lot. But what if you need the car later? It might be rented to another person and not be available to you. This can happen in Azure if no resources are available when you need to create the new VM.

3.3.5. Capacity Reservations

- In Azure we have a Capacity Reservation which reserves a certain size VM for your use even if you are not using it. Microsoft will charge you for the VM compute cost even when not using it.
- This would be similar to asking the rental company to park the car in their lot but not rent it to someone else in case you need it in the future. They will charge you for the car while it is sitting.

- There is an SLA with Capacity Reservations. Should you need the reserved VM and it not be available, Microsoft will provide a credit. This can happen if there is a hardware failure.
- With the rental car, this could be a tree falling on the car and then you need to use the car.

3.4. Pre-Sales Engineering – Planting the Seed

The pre-sales engineering team at TD SYNnex completes the activities that other pre-sales teams do such as creating BOMs and explaining product features; however, this is a small part of what we do. Most of our time is spent helping resellers and end users understand the cloud, how it relates to traditional IT services and how it can positively impact their business. The cloud is still new to many companies and there is significant fear, uncertainty, and doubt. It can take hours to months for a potential customer to feel confident with a migration, but the payoff for TD SYNnex and our reseller could be enormous.



An alternative way to consider, the Cloud Engineering teams efforts is as a farmer or Johnny Appleseed:

- **Selecting the Orchard Site** - choosing fertile ground is crucial for planting apple trees, but selecting the right cloud and understanding the customer needs is essential for cloud services. There are many clouds so the customer must feel confident when choosing the cloud upon which they will focus.
- **Planting the Seeds** - sowing seeds is akin to sparking interest with potential customers, overcoming fears, and planting the idea of how Microsoft Cloud can benefit their business.
- **Nurturing the Saplings** - providing continuous support and education about the cloud services is like watering and caring for the young trees, ensuring they grow strong. Forgetting about the saplings will cause them to wither or at least stunt their growth.
- **Pruning for Growth** - customizing solutions to fit customer needs helps trim away unnecessary costs or services, much like pruning a tree to promote healthy growth.
- **Protecting from Pests** – helping the customer understand potential cyber threats and how to combat them with robust security measures is as important as protecting apple trees from pests and disease.
- **Harvesting the Apples** - when the customer is ready, the cloud services can be integrated into the customer's business and yield positive results, it's time to harvest, reflecting successful sales and customer satisfaction.
- **Selling the Fruit** - the cloud is large and complex. The sales team needs engineering at their side for the final sale of cloud services. After demonstrating the value, is like selling ripe apples from the trees, which benefits both the reseller and the end user.

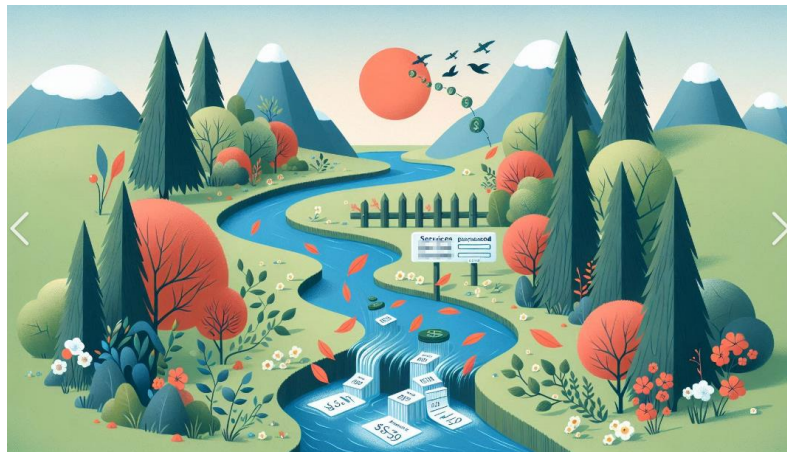
- **Replanting for the Future** – what was valuable today may not be tomorrow. Encouraging long-term relationships and repeat business ensures a sustainable sales cycle, much like planting new trees for future harvests. The engineering team must be constantly improving their knowledge and highlighting better solutions for our company to remain relevant.

3.5. Azure Costs

Azure costs are emitted at varying intervals during the month. Each resource may emit costs at different intervals and at different levels of granularity.

Generally, all costs have been emitted by the fifth day of the month following the billing month and normally no more than one day of billing may appear in the following month. Occasionally Microsoft may bill for services consumed in prior periods but not previously billed (this is very uncommon).

For these reasons, we should always assume that any costs observed are estimates until the invoice has been issued.



We often describe the emission of costs as a stream:

- Microsoft Azure Consumption is a stream starting at the beginning of the month (source) and ultimately dumping into the ocean or invoice at its mouth.
- All along the stream are resources such as trees and bushes that will drop their leaves (consumption costs) into the stream. There is no set interval when each one will drop their leaves and there may be more than one of the same type of tree (resource) along the banks.
- At any point along the stream (days during the month), we can stop at the stream and count the number of leaves of each type floating by, but this may not be all the leaves that will need to be counted.
- Since it takes time for all the leaves to float down the stream to the mouth, we can't just look at how many arrived at the mouth of the stream on the last day of the month. We need to wait for some that had dropped during the month to arrive at the month. This is the five days after the month for all costs to arrive previously mentioned.
- Sometime not all the leaves will arrive by the fifth day, so some may need to be counted in the next month.
- It is also possible for a leaf to get hung on a rock in the stream. Eventually it gets washed along and comes out of the mouth of the stream months later. Since that leaf was never counted it is not being double charged. It is very uncommon for leaves to get hung up.

3.6. Consumption Billing

Most Azure services are billed for what you use instead of a fixed fee every month. Even when a charge is provided as a monthly estimated charge, they are often still charged by the hour. Of course, there are exceptions where some resources have a fixed cost.



We can compare the Azure consumption billing to the electric service provided to your home or business.

- **Determining Need** – when initially contacting the electric utility they will ask some questions to determine how you plan to use electricity, what services or features might be useful, and they will often provide suggestions to save energy. The Azure pre-sale activities ask similar questions about your cloud needs.
- **Connect Service to Your House** – often the utility will connect the service to your house at no cost to you. You do not pay for the connection, only for what you use over the wires. This equates to an Azure subscription which has no cost itself.
- **Installing a Meter** – the utility will install a meter on your house to record the amount of electricity you use. Each month a meter reader stops by and reads the meter which helps determine the bill. All Azure services have meters that record the usage of Azure services. These meters are automatically read by Microsoft to determine the amount to be billed.
- **Your Electric Bill** – your electric bill reflects the cost of the electricity consumed in the prior month. It will go up and down depending on how much you use. This is the same for Azure; the more you consume the higher your bill.
- **Who Left the Lights On?** – the utility charges you for the electricity you consume. If you forget and leave the lights on all day, your costs will be higher. Most people will not expect the utility to credit them if they leave the lights on or if they ran an electric heater on high all month. This is analogous to leaving a virtual machine provisioned when not used or provisioning an expensive resource accidentally.

The services provided in Azure are consumable and Microsoft provides them as a utility. Just like your electric utility, Microsoft provides us tools to view and control our consumption; however, it is ultimately on us to manage our usage.

4. Definitions

Term	Definition
Stopped state	The virtual machine is allocated on a host but not running. Also called PoweredOff state or Stopped (Allocated).
Stopped (Deallocated) state	The virtual machine has released the lease on the underlying hardware. If the machine is powered off it is shown as Stopped (Deallocated).
Johnny Appleseed	An American pioneer nurseryman who introduced apple trees to large parts of PA, OH, IN, IL, WV, and ON. There is no evidence that he ever worked with the Microsoft cloud solutions.
MFA	Multi-factor authentication, or MFA, protects your applications by using a second source of validation before granting access to users. Common examples of multi-factor authentication include personal devices, such as a phone or token, or geographic or network locations
Sentinel	Microsoft Sentinel is a scalable, cloud-native security information and event management (SIEM) that delivers an intelligent and comprehensive solution for SIEM and security orchestration, automation, and response (SOAR).

5. Associated Documents

- [States and billing status - Azure Virtual Machines | Microsoft Learn](#)
- [What is Microsoft Sentinel? | Microsoft Learn](#)
- [Enable Microsoft Entra multifactor authentication - Microsoft Entra ID | Microsoft Learn](#)
- [Multifactor Authentication \(MFA\) | Microsoft Security](#)

6. Document Approval

This document has been approved by:

- Manager, Cloud Sales Engineering

7. Document Revision History

Number	Effective Date	Revision History	Author
1.0	6/5/2024	Initial Draft	David Bloom
1.1	6/11/2024	Added images	David Bloom
2.0	7/15/2024	Added consumption billing	David Bloom